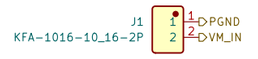
Power Supply Input



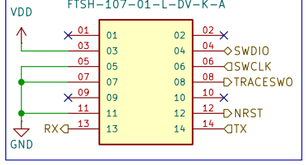
Phase Outputs



Power Supply Input
- High-power screw terminal block (Max 65A, 300V)
- Expects either raw wire or terminal connector tabs crimped on the wire across 6-26 AWG
- User will need to hook up an adapter to the connector used by the battery
- Note: 3D model's through-hole pins do NOT match the actual product/footprint (non-issue)
- Add terminal connector crimp kit to BOM

Phase Outputs
- High-power screw terminal block (Max 65A, 300V)
- Expects either raw wire or terminal connector tabs crimped on the wire across 6-26 AWG

Programming (SWD)



Programming (STDC14)

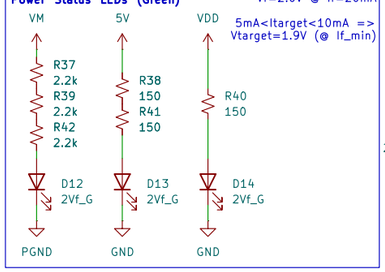
Expects STLINK-V3, so exposes pins through STDC14 connector

Uses MCU's internal VDD to determine if it's on

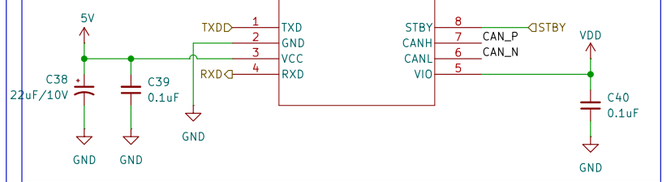
Power Status LEDs (Green)

Vf=2.0V @ If=20mA

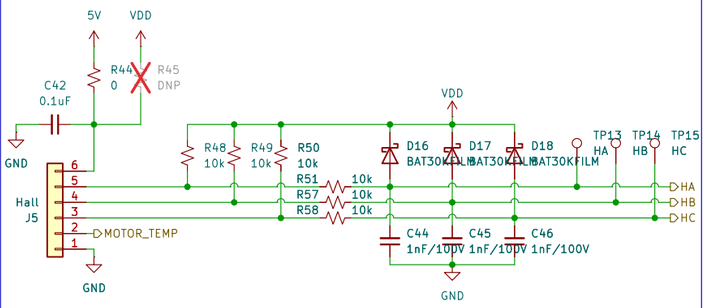
5mA<Itarget<10mA => Vtarget=1.9V (@ If_min)



CAN Transceiver



Hall Sensor Inputs



Hall Sensor Inputs

Assume VH required by motor hall sensors is 5V, but expose pad to switch to 3V3, if desired (XOR)

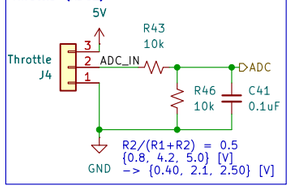
UM2896 5.4(?): Use CH1-3 of TIM2 (PA0, PA1, PA2) for HA, HB, HC (built-in XOR)

Optional MOTOR_TEMP pin if chosen motor provides winding temps (Note: This is different from the one in Control sheet)

ADC1_IN12 on PB1 used for MOTOR_TEMP

Notes:
- Only open-drain sensors are supported so added pull-up resistors and diodes to clamp voltage at 3V3 to protect MCU
- Active 5V push-pull hall sensors are NOT supported, as the pins couldn't be mapped such that they are 5V-tolerant

Throttle (ADC2)

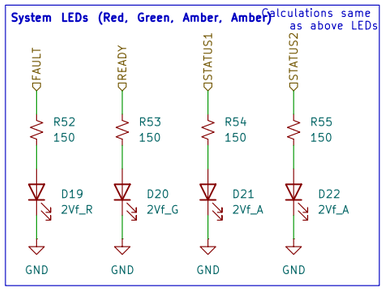


Throttle (JST-XH)

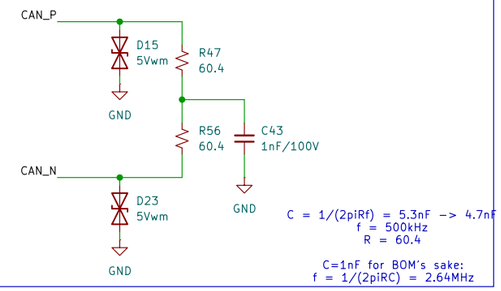
- Uses ADC2_IN5 (PC4)
- For bike's thumb throttle (wire cut off & stripped)
- Expects 5V ADC for full range
- Can also use an external pot

System LEDs (Red, Green, Amber, Amber)

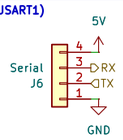
Calculations same as above LEDs



CANBus Split Termination



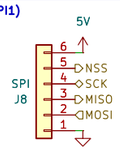
Serial (USART1)



Serial (JST-XH)

- Uses USART1_TX (PA9) & USART1_RX (PA10), which are 5V-tolerant
- For sk8's Flipsky VX1's receiver module (JST-PH, soldered to JST-XH connector)

SPI (SPI1)



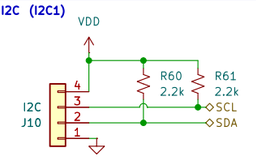
SPI (Optional)

Uses SPI1 (PA4-7), which are NOT 5V-tolerant

Only supports 3.3V TTL

May expect a rotary encoder (alternative to hall sensor)

I2C (I2C1)

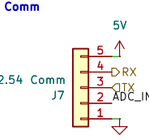


I2C (Optional)

Uses I2C1_SDA (PB7), I2C1_SCL (PA15)

May expect a display? Just exposing it to future-proof

2.54 Comm

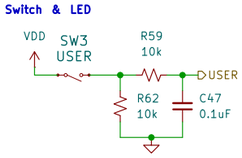


2.54 Comm (2.54mm Pitch Header)

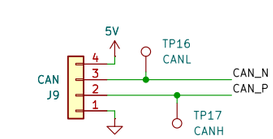
Serial and ADC exposed twice (once with their own JST connectors, and another while combined with regular male pins)

For debugging

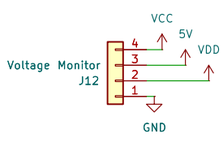
User Switch & LED



CANBus Connector



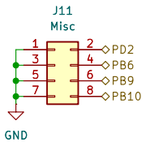
Voltage Monitor



Voltage Monitor

Most power-related signals exposed through test points near pins

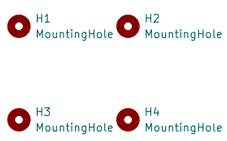
Misc. Header Pins



Misc. Header Pins

Four unused pins remain, so pair with GND and expose

Mounting Holes



Mounting Holes

Using non-plated through-holes (NPTH) sized for screw-in rubber bumpers

Add additional parts for standoffs to BOM

Author: Leopoldo Mendoza, Eleanor Wang
Electrium Mobility

Sheet: /Peripherals/
File: peripherals.kicad_sch

Title: ESC 2026 - Peripherals

Size: A3	Date: 2026-08-06	**Rev: v1**
KiCad E.D.A. 10.0.3		Id: 5/5